Kory Pritchett

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Summary

I design the curriculum and grading infrastructure that approximately 2,600 first-year engineering students learn on each semester at Purdue, from the GitHub Actions pipelines that deploy ENGR 131 to the Gradescope autograders that grade 30 of its assignments. I also build Python integrations against REST APIs and undocumented formats: a self-hosted Ollama summarization pipeline deployed in Docker, and a ProPresenter writer built on a reverse-engineered Protocol Buffer schema. That nonprofit suite cut a 4-5 hour weekly task to under 15 minutes with near-zero copy-paste errors.

Work Experience

Senior Curriculum Developer

Purdue University, First-Year Engineering (FYE) | October 2021 – Present

- Led the technical migration of ENGR 131 off fragmented per-section materials onto one version-controlled GitHub, Codespaces, and JupyterBook platform serving 20+ sections; the rewritten course shipped for Fall 2026 and is live.
- Built the GitHub Actions CI/CD pipelines that compile and deploy the course website, plus a 62-test prose-style checker that holds pages from several authors to one style guide.
- Wired Gradescope autograders to the course repository so 30 assignments grade automatically and a test fix deploys when it merges; extended the generic-grader Python framework to grade Excel and MATLAB submissions.
- Hardened the Gradescope autograder grading ~2,600 first-year engineering students each semester: rebuilt the SSH deploy-key pipeline to fail loudly instead of silently shipping broken grading, rewrote the late-penalty engine to read deadlines from Gradescope submission metadata, and authored the technical docs the TA team uses.
- Developed Python and MATLAB curricula (videos, assignments, assessments, modular projects) for approximately 2,600 students per semester, and administered course content and learner access through D2L Brightspace, Gradescope, and GitHub.
- Cut roughly a week of manual grade analysis and reporting to about 15 minutes with Python automation that surfaces the outcomes and modules needing attention.

Technology Lead

First United Methodist Church, West Lafayette, IN | 2025 – Present

- Cut a 4-5 hour weekly content process to under 15 minutes and took copy-paste errors from moderate to near zero by building a Python monorepo that generates bulletins and slides.
- Integrated Gmail and Planning Center through their REST APIs with a unified Python runner that handles OAuth2 flows and caches API responses to stay within rate limits.
- Moved newsletter summarization off a paid cloud model onto a self-hosted Ollama LLM deployed in Docker, keeping data on-prem at zero per-call cost.
- Reverse-engineered ProPresenter 7’s undocumented Protocol Buffer schema (45+ message types) and wrote a native Python writer that generates slides without ever opening the app.
- Administered the Microsoft 365 tenant and ran livestream production, ProPresenter, Allen & Heath audio mixing, and AV systems.

Math Teacher

West Lafayette Jr./Sr. High School | *August 2017 – August 2021*

- Taught grades 7-12 math to 100+ students per semester, in person and virtually.
- Led the department's adoption and deployment of Canvas LMS, coordinating the rollout across courses and helping colleagues migrate course materials into new digital workflows.

Skills

Languages & Tools

- Python, TypeScript/Bun, Git/GitHub, GitHub Actions, CI/CD pipelines, Docker, Linux administration

Automation & Integration

- REST API orchestration (Gmail, scheduling/planning tools), self-hosted local LLMs (Ollama), Protocol Buffers, modular Python architectures

LMS & Assessment

- Brightspace (D2L), Gradescope, Canvas, Jupyter Book, MyST, GitHub Pages, GitHub Codespaces

Infrastructure & AV

- Proxmox, LXC, UniFi VLAN segmentation, Tailscale, Cloudflare Tunnel, Dante, ProPresenter, Allen & Heath (SQ, Qu), Bitfocus Companion

Data

- pandas, NumPy, Matplotlib, Plotly

Projects

Self-Healing Homelab Infrastructure (TypeScript/Bun · Proxmox · Docker · Linux)

- Built an always-on TypeScript/Bun watchdog after a real 12-hour outage: it senses internet loss three independent ways (ICMP, DNS, HTTPS) and power-cycles the modem through Home Assistant's REST API.
- Operated a multi-host Proxmox cluster of VMs, LXC containers, and Docker services on Debian and Ubuntu Server, with backups to network-attached storage and UPS-triggered graceful shutdown across the fleet.
- Designed UniFi networks with VLAN segmentation, firewall rules, and encrypted remote access, exposing public services through a zero-trust Cloudflare Tunnel.

Meeting & Interview Summarization (Self-Hosted LLM)

- Built a local LLM pipeline that summarizes meetings and interviews on self-hosted models, so the source content stays on local infrastructure.

Education

M.A. Science Education

Ball State University, Muncie, IN | *May 2017*

B.S. Physics

Purdue University, West Lafayette, IN | *May 2015*